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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Computer Science & Engg.) (CBCS) Degree Examinations

May - June 2024

23CSE202 – BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five** full questions choosing **One** full question from each **Unit**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) What is Big Data Analytics? Why is Big Data Analytics important?	8	L*2	1	1
b) What is NoSQL? Explain the types of NoSQL databases with example.	8	L2	1	1
c) Discuss the characteristics of the data.	4	L1	1	1
2. a) What is Bigdata? Explain the characteristics of Bigdata.	8	L2	1	1
b) What is sharding? Explain with examples.	6	L2	1	1
c) What is NoSQL? Give the difference between SQL and NoSQL.	6	L2	1	1

Unit – II

3. a) Differentiate between insert () and update () methods with the help of MongoDB queries. What is use of upsert and mutli options in update () method? Explain.	8	L3	2	1
b) Discuss the CAP theorem in detail.	6	L2	2	1
c) What is Replication? How replication works in MongoDB?	6	L2	2	1
4. a) Outline the various datatypes in MongoDB.	8	L2	2	1
b) Consider the collection "Students" with student documents that contains fields namely, roll number, name, grade and hobbies.				
a. Find the number of documents in the Students collection wherein the Grade is VII.				
b. Sort the documents from the Students collection first on Grade in ascending order and then on Hobbies in descending order.				
c. Sort the documents from the Students collection and skip the first document from the output.				
d. Display the last 2 documents from the Students collection.	8	L3	2	1
c) What is MongoDB? Why is it used?	4	L1	2	1

Unit – III

5. a) What is Hadoop? Why is Hadoop used? Explain.	6	L2	3	1
b) Describe briefly HDFS Daemons.	8	L2	3	1
c) Outline the main stages of Hadoop MapReduce data processing.	6	L2	3	1
6. a) How weather data is analyzed with Hadoop for finding the maximum temperature? Explain with mapper and reducer code.	10	L3	3	1
b) With a neat diagram, Explain Hadoop 2 YARN Architecture.	10	L2	3	1

Unit – IV

7. a) Discuss the following.				
a. Iterative Operations on MapReduce.				
b. Interactive Operations on MapReduce.				
c. Data Sharing using Spark RDD.				
d. Iterative Operations on Spark RDD.	8	L2	4	1
b) What is HBase? Outline features of HBase.	6	L2	4	1
c) Discuss the role of Apache Scoop and Apache Zookeeper.	6	L2	4	1
8. a) With a neat diagram explain the components of Spark application.	10	L2	4	1
b) Discuss Hadoop ecosystem briefly.	10	L2	4	1

Unit – V

9. a) What is Hive? Give the difference between SQL and HiveQL.	6	L2	5	1
b) Discuss the LOAD and STORE operators in Pig.	4	L3	5	1
c) With a neat diagram explain PIG architecture.	10	L2	5	1
10. a) Explain the different datatypes in Hive and commands for loading data into Hive.	10	L2	5	1
b) Write a word count program using Pig Latin.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
